CS1 Week 6 Annotated

Transfer functions: Poles and Zeros

Attention

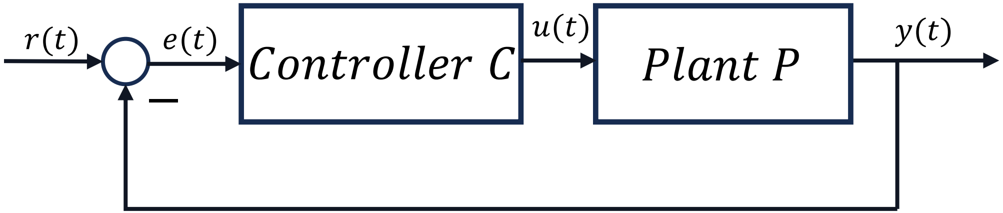
- ETH no longer provides web hosting.
- All my exercise class materials have been moved to GitHub Pages.
- You can now find them under the following link:

https://kissanv.github.io

Or you can scan the QR-Code:

Recap Last week

Motivation



system modeling → **Plant P** (*Ferrari 812*) → **Linearization** →

state-space representation

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t)$$

Analysis ↓

Transfer Functions → **Synthesis** → **Controller C**

Time response:

$$y = y_{IC} + y_F = Ce^{At}x_0 + C\int_0^t e^{A(t-\tau)}B\,u(\tau)d\tau + Du(t)$$

I.C. response ***Forced response***

$$x_{IC}(t) = e^{\tilde{A}t}x(0) = diag\left(e^{\lambda_1 t}, \dots, e^{\lambda_n t}\right)$$
$$y_{IC}(t) = \tilde{C}e^{\tilde{A}t}x(0) = \Sigma_{i=1}^{n} c_i e^{\lambda_i t}\tilde{x}_{0,i}$$

$$y(t) = \Sigma u(t)$$
$$Y(s = G(s)U(s)$$
$$\mathcal{L}\{\cdot\}$$

Time Response

- We're still dealing with LTI systems: $\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t) \end{cases}$
- We solved the ODE by splitting into initial and forced response

$$\left.\begin{array}{l} \text{Time response of a first-order LTI system:} \\ x(t) = e^{at}x_0 + \int_0^t \phi(t-\tau)bu(\tau)\, d\tau, \\ y(t) = ce^{at}x_0 + c\int_0^t \phi(t-\tau)bu(\tau)\, d\tau + du(t), \\ \text{with } \phi(t-\tau) := e^{a(t-\tau)} \end{array}\right\}\ \text{difficult to compute}$$

- We want an easier way of describing the I/O relation

→ Transfer function

Transfer Function

- Transfer function gives the relation between Y and U: $\underbrace{G(s) = \frac{Y}{U} \longleftrightarrow Y = GU}_{s\text{-domain}}$
- We derived the general TF for LTI state-space:

$$\boxed{G(s) = C(sI - A)^{-1}B + D}$$

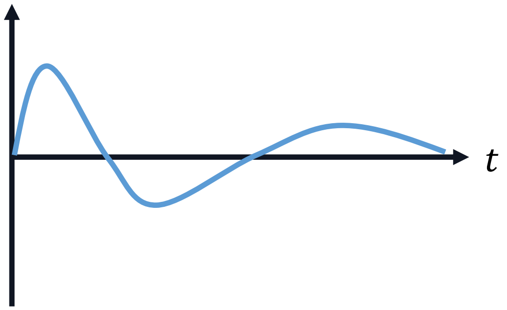
- To find the inverse $(sI - A)^{-1}$ we use Cramer's rule:

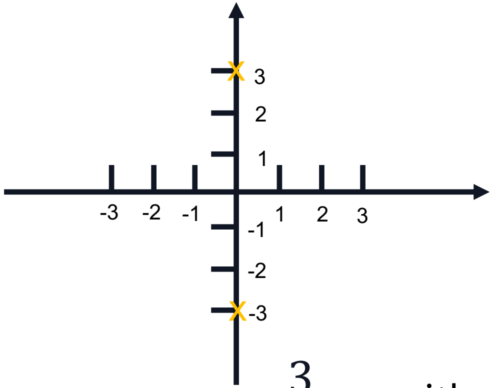
$$\boxed{M^{-1} = \frac{adj(M)}{det(M)} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}}$$

- $G(s)$ is defined in the s-domain but describes behavior in the time domain
- Remember the steady-state output to the input $u(t) = e^{st}$ is: $y_{ss} = G(s)\, e^{st}$

s-Domain

With the TFs we start to represent systems in a new domain. So instead of characterizing how systems behave in the time domain, i.e. $y(t) = \cdots$, we start to look at how it behaves in the s-domain, i.e. $Y(s) = G(s)U(s)$. But what exactly is the s-domain? ($s = \sigma + j\omega$)





$$e^{-0.8t}\sin(3t)$$

$$\mathcal{L}\{e^{-0.8t}\sin(3t)\} = \frac{3}{s^2+9}$$

with poles at $s = \pm 3j$

For now you can think that in the s-domain functions or systems are represented by their poles (and zeros) plotted in the *complex* plane.

Quiz 1

What are the transfer functions of an (1) integrator and a (2) differentiator?

A. $G_1(s) = s, \quad G_2(s) = \frac{1}{s}$

B. $G_1(s) = \frac{1}{s}, \quad G_2(s) = s$

C. They don't have transfer functions

D. $G_1(s) = \frac{s+1}{s}, \quad G_2(s) = \frac{1}{s+1}$

Quiz 1

What are the transfer functions of an (1) integrator and a (2) differentiator?

A. $G_1(s) = s, \quad G_2(s) = \frac{1}{s}$

B. $G_1(s) = \frac{1}{s}, \quad G_2(s) = s$

C. They don't have transfer functions

D. $G_1(s) = \frac{s+1}{s}, \quad G_2(s) = \frac{1}{s+1}$

Integrator: $u(t) \longrightarrow \boxed{\int} \longrightarrow y(t)$

For an input $u(t) = e^{st}$, the output will be $y(t) = \frac{1}{s} e^{st}$

Differentiator: $u(t) \longrightarrow \boxed{\frac{d}{dt}} \longrightarrow y(t) = \frac{du(t)}{dt}$

For an input $u(t) = e^{st}$, the output will be $y(t) = s e^{st}$

Controllable Canonical Form

- We know how to go from state-space to Transfer function, for the other way we use the canonical form
- For a general TF $G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} + d$

A minimal realization of G(s) in **rational function form** is:

$$\left[\begin{array}{c|c} A & b \\ \hline c & d \end{array}\right] = \left[\begin{array}{ccccc|c} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & & 1 & 0 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} & 1 \\ \hline b_0 & b_1 & b_2 & \dots & b_{n-1} & d \end{array}\right]$$

$$adj\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} ei - fh & ch - bi & bf - ce \\ fg - di & ai - cg & cd - af \\ dh - eg & bg - ah & ae - bd \end{bmatrix}$$

Quiz 2

Box 1: Questions 8, 9, 10

You are given a linear time-invariant system in state-space form.

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} x(t)$$

Question 8 *Choose the correct answer. (1 Point)*

A. $G(s) = \frac{3+2s}{s^2(s+1)}$

C. $G(s) = \frac{3-2s}{s(s+1)}$

B. $G(s) = \frac{1}{s+1}$

D. $G(s) = \frac{3}{s(s+1)}$

$$adj\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} ei-fh & ch-bi & bf-ce \\ fg-di & ai-cg & cd-af \\ dh-eg & bg-ah & ae-bd \end{bmatrix}$$

Quiz 2

Box 1: Questions 8, 9, 10

You are given a linear time-invariant system in state-space form.

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} x(t)$$

Question 8 *Choose the correct answer. (1 Point)*

A. $G(s) = \frac{3+2s}{s^2(s+1)}$

C. $G(s) = \frac{3-2s}{s(s+1)}$

B. $G(s) = \frac{1}{s+1}$

D. $G(s) = \frac{3}{s(s+1)}$

$$G(s) = C\,\frac{\underline{adj(sI-A)}}{det(sI-A)}\,B$$

$$det(sI-A) = \begin{vmatrix} s & -1 & -2 \\ 0 & s & -3 \\ 0 & 0 & s+1 \end{vmatrix} = s\begin{vmatrix} s & -3 \\ 0 & s+1 \end{vmatrix}$$

$$= s^2(s+1)$$

$$G(s) = \frac{1}{s^2(s+1)}\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}\underbrace{\begin{bmatrix} s(s+1) & s+1 & 3+2s \\ 0 & s(s+1) & 3s \\ 0 & 0 & s^2 \end{bmatrix}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}_{\begin{pmatrix} 3+2s \\ 3s \\ s^2 \end{pmatrix}}$$

$$G(s) = \frac{s^2}{s^2(s+1)} = \frac{1}{s+1}$$

Course Schedule

	Subject	Week
Modeling	Introduction, Control Architectures, Motivation	1
	Modeling, Model examples	2
	System properties, Linearization	3
Analysis	Analysis: Time response, Stability	4
	Transfer functions 1: Definition and properties	5
	Transfer functions 2: Poles and Zeros	6
	Proportional feedback control, Root Locus	7
	Time-Domain specifications, PID control, Computer implementation	8
	Frequency response, Bode plots	9
	The Nyquist condition, Time delays	10
Synthesis	Frequency-domain Specifications, Dynamic Compensation, Loop Shaping	11
	Time delays, Successive loop closure, Nonlinearities	12
	Describing functions	13
	Intro to Uncertainty and Robustness	14

Today

1. Different TF Notations
2. System Response
3. Poles & Zeros

1. Different TF Notations

Partial Fraction Expansion

Note strictly proper: $\frac{s^m}{s^n}, n > m$

- Recall the ***General Form*** of transfer function:

$$\boxed{G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} + d}$$

Note: useful to switch between state-space and TF with canonical form

- In some cases, it's helpful to represent TF in different forms, like

the ***Partial Fraction Expansion:***

$$\boxed{G(s) = \frac{r_1}{s - p_1} + \frac{r_2}{s - p_2} + \cdots + \frac{r_n}{s - p_n} + r_0}$$

r_i: residues

Note: useful to use inverse Laplace to get the time response of each pole

$$y(t) = r_1 e^{p_1 t} + r_2 e^{p_2 t} + \cdots + r_n e^{p_n t}$$

How to determine residues

1. Partial Fraction Decomposition (Analysis I/II)
2. Cover-up method: For non-repeating poles p_i: $r_i = \lim_{s \to p_i} (s - p_i)\, G(s)$

For repeating pole p_i of order m: $r_i = \frac{1}{(m-1)!} \lim_{s \to p_i} \frac{d^{m-1}}{ds^{m-1}} [(s - p_i)^m G(s)]$

Root-Locus- & Bode-Form

These are also other forms to represent the TF, recall from last week:

- **Poles $\boldsymbol{p_i}$** are the roots of the *Denominator* of $G(s)$
- **Zeros $\boldsymbol{z_j}$** are the roots of the *Nominator* of $G(s)$

- ***Root Locus Form:***

$$\boxed{G(s) = k_{rl}\frac{(s-z_1)(s-z_2)\dots(s-z_m)}{(s-p_1)(s-p_2)\dots(s-p_n)}}$$

Used for Root Locus

Note: we can directly see all poles & zeros

- ***Bode Form:***

$$\boxed{G(s) = \frac{k_{bode}}{s^q}\frac{\left(\frac{s}{-z_1}+1\right)\left(\frac{s}{-z_2}+1\right)\dots\left(\frac{s}{-z_m}+1\right)}{\left(\frac{s}{-p_1}+1\right)\left(\frac{s}{-p_2}+1\right)\dots\left(\frac{s}{-p_{n-q}}+1\right)}}$$

Used for Bode Plots

Example $G(s) = \frac{4(s+3)}{s^2+3s+2}$

zeros: $z_1 = -3$

poles: $s^2+3s+2 = (s+2)(s+1) = 0$

$\hookrightarrow p_1 = -1,\ p_2 = -2$

1. Root-Locus: $G(s) = 4\frac{(s+3)}{(s+2)(s+1)}$

2. Bode-Form:

$$G(s) = 4\frac{3(s/3+1)}{2(s/2+1)(s+1)}$$

$$= 6\frac{(s/3+1)}{(s/2+1)(s+1)}$$

3. Partial Fraction Expansion:

$$G(s) = \frac{r_1}{s+2} + \frac{r_2}{s+1}$$

cover-up method:

$$r_1 = \lim_{s \to -2} \cancel{(s+2)}\, 4\frac{(s+3)}{\cancel{(s+2)}(s+1)} = 4 \cdot \frac{1}{-1}$$

$$= -4$$

$$r_2 = \lim_{s \to -1} \cancel{(s+1)}\, 4\frac{(s+3)}{(s+2)\cancel{(s+1)}} = 4\frac{2}{1}$$

$$= 8$$

$$\hookrightarrow G(s) = \frac{-4}{s+2} + \frac{8}{s+1}$$

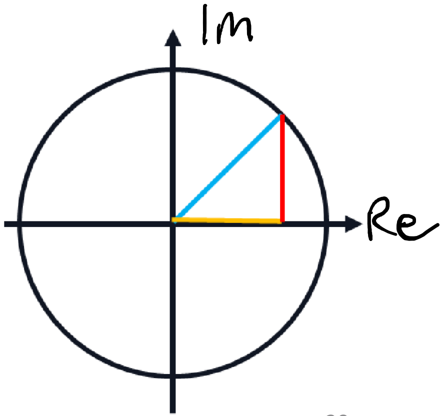
2. System Response

Recap Complex Numbers

$j^2 = -1$

- Every complex number $z \in C$ can be represented by its **magnitude** and **phase**
- Geometrically, $z = a + jb$ corresponds to a vector in the complex plane

1. *Computing magnitude:* $|z| = \sqrt{a^2 + b^2}$
2. *Computing phase:* $\angle z = \tan^{-1}(\frac{b}{a})$
3. *Vector interpretation:* The difference $(z_1 - z_2)$ represents the vector from point z_2 to point z_1 in the complex plane



- For two complex numbers z_1 & z_2 following rules hold:

$$|z_1 z_2| = |z_1| \cdot |z_2|, \qquad |z_1/z_2| = \frac{|z_1|}{|z_2|}$$

$$\angle(z_1 z_2) = \angle z_1 + \angle z_2, \qquad \angle(z_1/z_2) = \angle z_1 - \angle z_2$$

Sinusoidal Response

If Input is:	$u(t) = \cos(\omega t)$
Output given by:	$y(t) = M\cos(\omega t + \phi)$
with	$M = \lvert G(j\omega)\rvert, \phi = \angle G(j\omega)$

Example: Given $u(t) = \cos(t)$ let $G(s) = 2\frac{s+1}{(s+2)(s+1+j)(s+1-j)}$

Goal: Compute the time response of the system:

$$G(j) = 2\frac{1+j}{(2+j)(1+2j)}$$

1. Compute $M = |G(j)|$ $\quad |G(j)| = 2\frac{\sqrt{2}}{\sqrt{5}\sqrt{5}} = \frac{2\sqrt{2}}{5}$

2. Compute $\phi = \angle G(j)$ $= \angle(2) + \angle(1+j) - \angle(2+j) - \angle(1+2j)$

$$= 0^\circ + 45^\circ - \arctan(1/2) - \arctan(2) = -45^\circ$$

$$y_{ss} = \frac{2\sqrt{2}}{5}\cos(t - 45^\circ)$$

Graphical Approach

$$G(s) = k_{rl}\frac{(s-z_1)(s-z_2)\dots(s-z_m)}{(s-p_1)(s-p_2)\dots(s-p_n)}$$

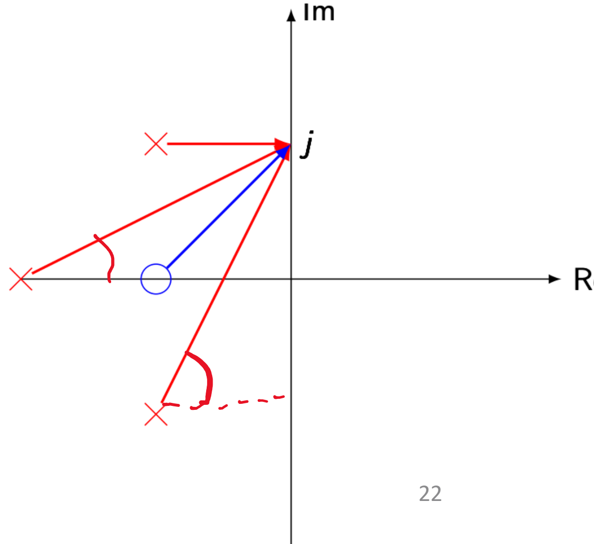
Each term $(s-z_i)$ and $(s-p_i)$ is a complex vector from zeros and poles to s; using the **magnitude and phase rules**, the total $|G(s)|$ and $\angle G(s)$ result from the product of lengths and sum of angles.

Root-locus gain

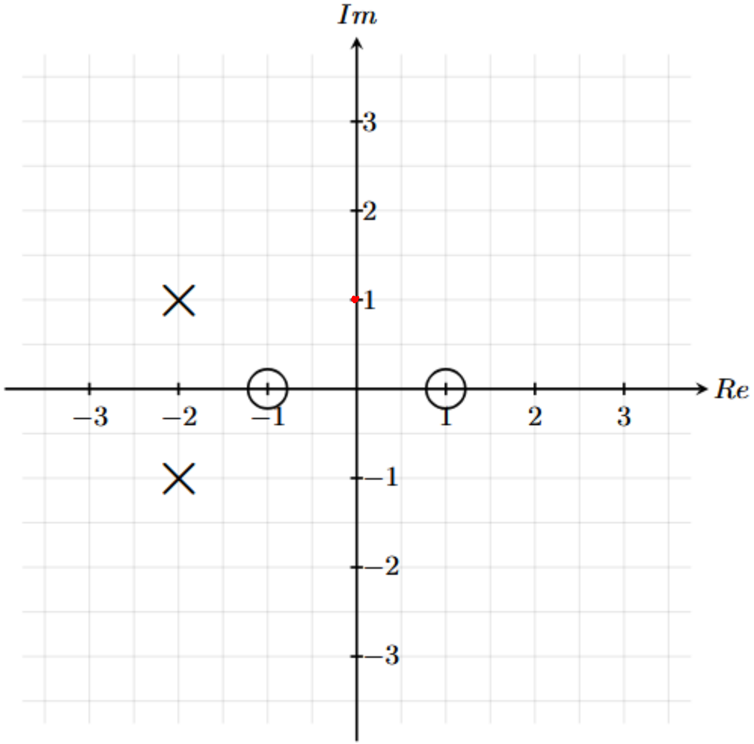
Magnitude: $|G(j)| = \frac{\sqrt{2}}{\sqrt{5}\cdot\sqrt{5}\cdot 1}\cdot 2 = \frac{2\sqrt{2}}{5}$

Argument: $\angle G(j) = 45^\circ - 0^\circ - \arctan(1/2)$
$- \arctan(2) = -45^\circ$

$y_{ss} = \frac{2\sqrt{2}}{5}\cos(t-45^\circ)$



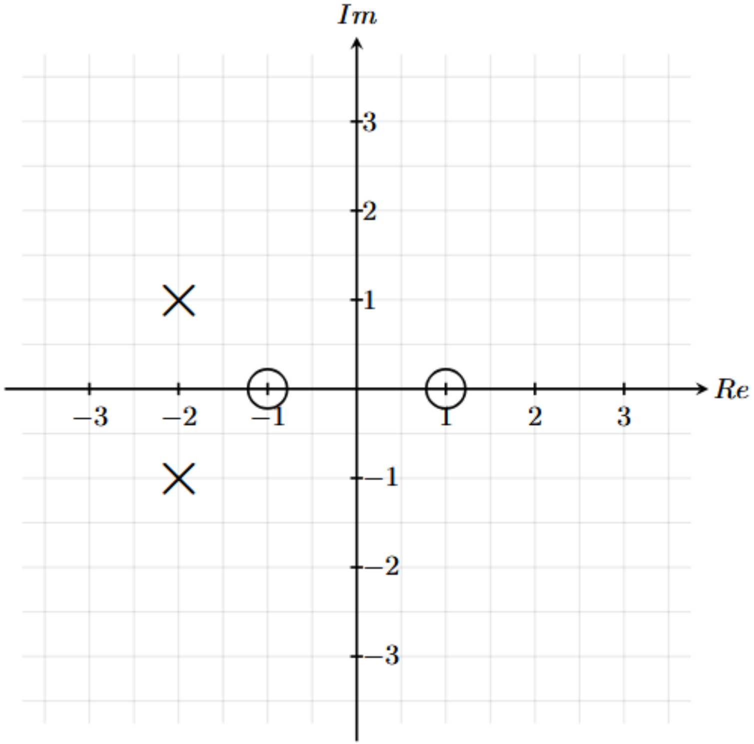
Quiz



What is the magnitude $|G(j)|$ of $G(s = j)$?

A. $\frac{1}{2\sqrt{2}}$ **B.** $\frac{1}{4}$ **C.** $\frac{1}{2}$ **D.** $\frac{1}{\sqrt{2}}$

Quiz



What is the magnitude $|G(j)|$ of $G(s = j)$?

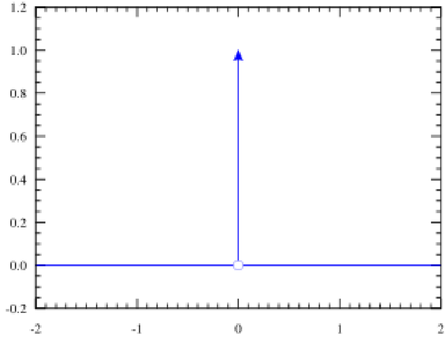
A. $\frac{1}{2\sqrt{2}}$

B. $\frac{1}{4}$

C. $\frac{1}{2}$

D. $\frac{1}{\sqrt{2}}$

Impulse Response



E.g. input: $u(t) = \delta(t) = \begin{cases} \infty & t = 0 \\ 0 & otherwise \end{cases}$ $D = 0$ and $x(0) = 0$

Forced response: $y_F(t) = y_{\text{imp}}(t) = C\int_0^t e^{A(t-\tau)}B\,u(\tau)d\tau = Ce^{At}B$

This is equivalent to the initial condition response if $x(0) = B$

Another way to compute the forced response is with the Laplace transform:

$\mathcal{L}\big(\delta(t)\big) = 1 \longrightarrow Y(s) = G(s)U(s) = Y(s) = G(s)\mathcal{L}\big(\delta(t)\big) = G(s) \cdot 1$

$\longrightarrow y_{\text{imp}}(t) = \mathcal{L}^{-1}\big(G(s)\big)$

If given partial fraction expansion of a TF: $G(s) = \frac{r_1}{s-p_1} + \frac{r_2}{s-p_2} + \cdots + \frac{r_n}{s-p_n} + r_0$

$$y_{imp}(t) = \mathcal{L}^{-1}\big(G(s)\big) = r_1 e^{p_1 t} + r_2 e^{p_2 t} + \cdots + r_n e^{p_n t}$$

$\mathcal{L}\{e^{at}\} = \dfrac{1}{s-a}$

→ We note that each pole generates an exponential in the time domain!

3. Poles & Zeros

Effects of Poles

$$G(s) = C\frac{adj(sI - A)}{det(sI - A)}B + D$$

The poles of a TF are calculated by setting the **denominator** to zero

- Why are we interested in poles of a TF?

→ the denominator of $G(s)$ is $det(sI - A)$

→ to get the poles we solve $det(sI - A) = 0$

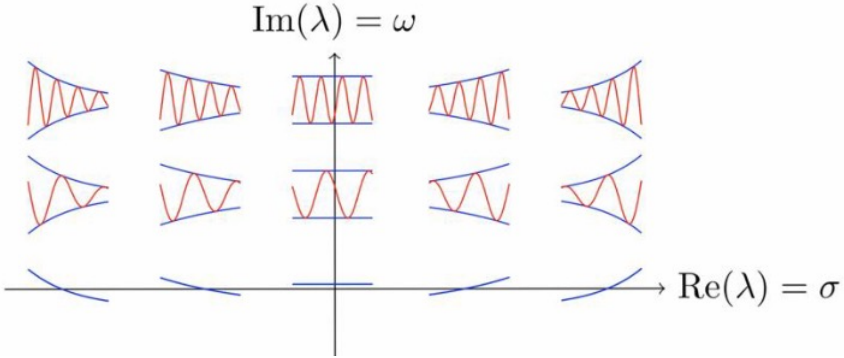
→ this look like an EV problem? *Poles = Eigenvalues of A!*

- *Time response*: solution $x(t)$ describing how the state evolves over time

$$x(t) = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2 + \cdots$$

- Poles are directly connected how a system behaves!

Effects of Poles



- Can be complex → appear in conjugate pairs and have an oscillatory effect
- Recall stability discussion from Week 4:

A linearized, diagonalized system with A, B, C, D matrices is called

- **Lyapunov stable** if $Re(\lambda_i) \leq 0 \ \forall i$
- **Asymptotically** stable if $Re(\lambda_i) < 0 \ \forall i$
- **Unstable** if $\exists \ Re(\lambda_i) > 0 \ \forall i$

A linearized system with non-diagonizable A matrix is called

- **Lyapunov stable** if $Re(\lambda_i) \leq 0 \ \forall i$ and there are no repeated eigenvalues with $Re(\lambda_i) = 0$

Effects of Zeros

- The zeros of a TF are calculated by setting the **nominator** to zero
- To look at the effects of zeros consider the Transfer Function $G(s)$ containing a zero:

$$G(s) = \left(\frac{s}{-z} + 1\right)\hat{G}(s) = \hat{G}(s) + \frac{s}{-z}\hat{G}(s)$$

Let $\hat{y}(t)$ be the response of $\hat{G}(s)$. Then $\dot{\hat{y}}(t)$ is the response of $\overbrace{s\hat{G}(s)}^{\text{derivative in Laplace:}}$

$$\longrightarrow Y(s) = G(s)U(s) = \left(\hat{G}(s) + \frac{s}{-z}\hat{G}(s)\right)U(s)$$

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\{\hat{G}(s)U(s)\} + \frac{1}{-z}\mathcal{L}^{-1}\{s\hat{G}(s)U(s)\}$$

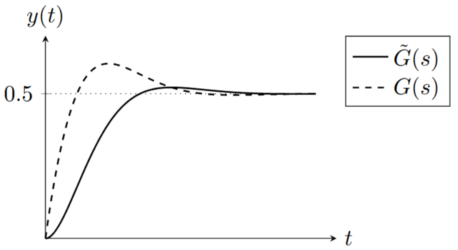
$$y(t) = \hat{y}(t) + \frac{1}{-z}\dot{\hat{y}}(t)$$

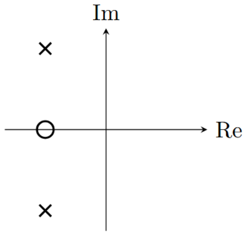
→ zeros add a derivative term to the output
→ *The closer the zero is to the origin, the <u>stronger its effect</u>*
→ this means we usually have an anticipatory effect but we will discuss two cases…

Zeros as Derivative Actions

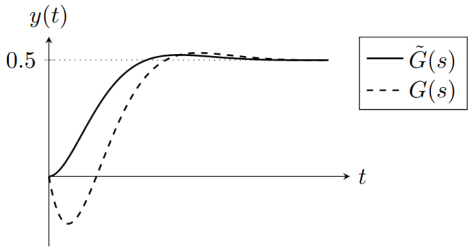
$$\boxed{y(t) = \hat{y}(t) + \frac{1}{-z}\dot{\hat{y}}(t)}$$

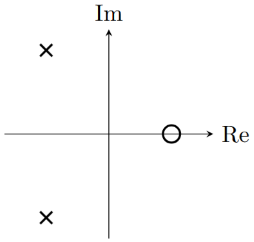
Minimum Phase Zero





Non Minimum Phase Zero





→ ***Minimum-Phase Zero:*** $Re(z) \leq 0$: adds an anticipatory effect

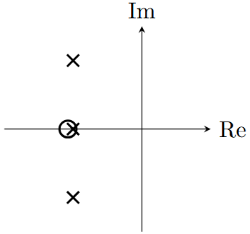
→ ***Non-Minimum-Phase Zero:*** $Re(z) > 0$: adds "negeative" derivative action

→ zeros affect the output but they <u>can't</u> influence stability

Pole-Zero Cancellation

We would like to analyse what happens when a pole and a zero are close to each other, and or if we would cancel them in the TF. Consider the following TF

$$G(s) = \frac{s+1+\varepsilon}{(s+1)(s+1+j)(s+1-j)}$$



Remember that looking at the partial fraction expansion, we actually see how much a pole is present in the response. So let's compute the residues:

$$r_1 = \lim_{s\to -1}(s+1)G(-1) = \varepsilon$$

$$r_2 = \lim_{s\to -1-j}(s+1+j)G(-1-j) = \frac{j-\varepsilon}{2}$$

$$r_3 = \lim_{s\to -1+j}(s+1-j)G(-1+j) = \frac{-j-\varepsilon}{2}$$

$$\longrightarrow \boxed{G(s) = \frac{\varepsilon}{s+1} + \frac{0.5(j-\varepsilon)}{s+1+j} - \frac{0.5(j+\varepsilon)}{s+1-j}}$$

Pole-Zero Cancellation

$$\boxed{G(s) = \frac{\varepsilon}{s+1} + \frac{0.5(j-\varepsilon)}{s+1+j} - \frac{0.5(j+\varepsilon)}{s+1-j}}$$

- Remember that we chose the zero being arbitrarily close to the pole. We can see now that when **we move the zero closer to the pole**, the residue of this pole, **it's effect on the response, gets smaller**, until it gets 0 eventually.

- This is fine as long as we do not cancel any unstable poles.
- **DO NOT CANCEL UNSTABLE POLES!** (System blows up)

Initial & Final Value Theorem

- We can use the following trick to compute the initial/final value of $y(t)$

Initial Value Theorem:

$$\lim_{t\to 0} y(t) = \lim_{s\to\infty} s\,Y(s) = \lim_{s\to\infty} s\,G(s)U(s)$$

Final Value Theorem:

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} s\,Y(s) = \lim_{s\to 0} s\,G(s)U(s)$$

We can apply this to derivatives:

$$\lim_{t\to 0} \dot{y}(t) = \lim_{s\to\infty} s\left[\mathcal{L}\big(\dot{y}(t)\big)\right] = \lim_{s\to\infty} s^2\,Y(s) = \lim_{s\to\infty} s^2\,G(s)U(s)$$

Old Exam Question

x: pole / o: zero

Problem: The step response plot of four different 2nd order systems is shown in Figure 19. The poles and zeros plots are given in Figure 20.

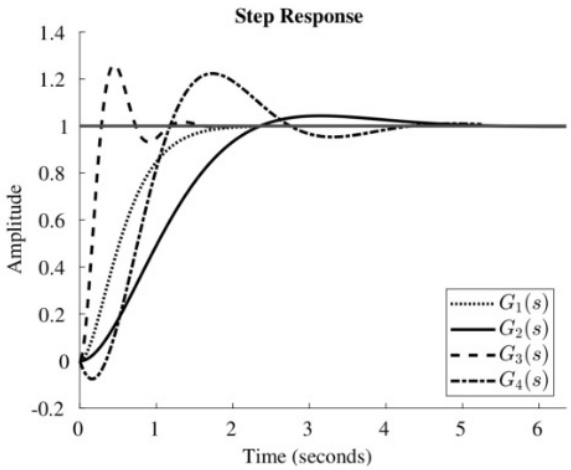


Figure 19: Step reponse plot of four different 2nd order systems. The horizontal line with amplitide equals 1 shows the unit step function.

Q35 **(1.5 Points)** Mark the correct combinations of poles and zeros plots in Figure 20 with the corresponding 2nd order step response in Figure 19.

Transfer functions	A	B	C	D
$G_1(s)$				
$G_2(s)$				
$G_3(s)$				
$G_4(s)$				

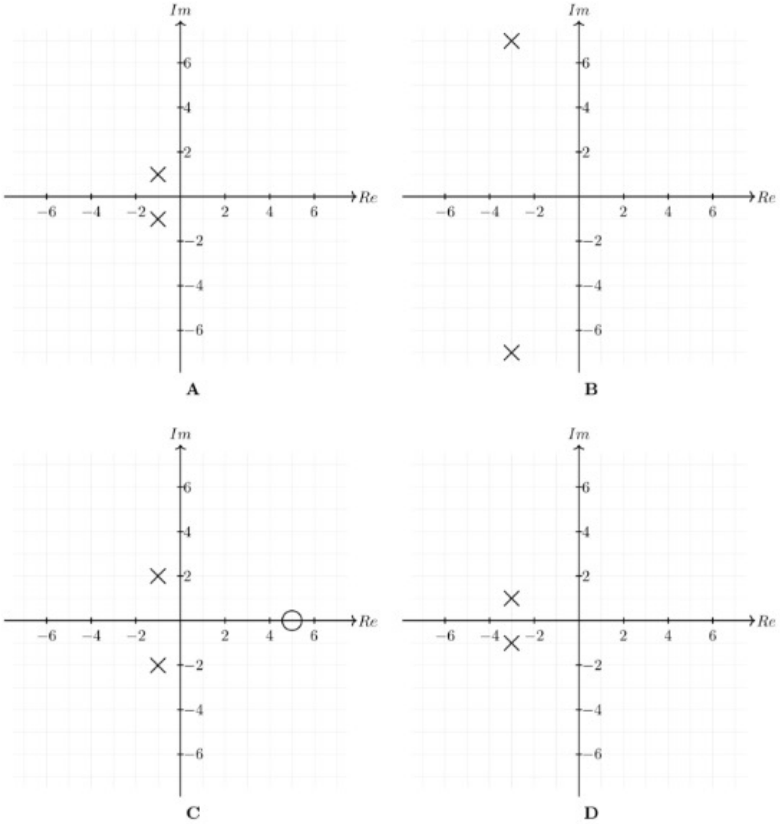


Figure 20: Poles (×) and zeros (○)

Old Exam Question

x: pole / o: zero

Problem: The step response plot of four different 2nd order systems is shown in Figure 19. The poles and zeros plots are given in Figure 20.

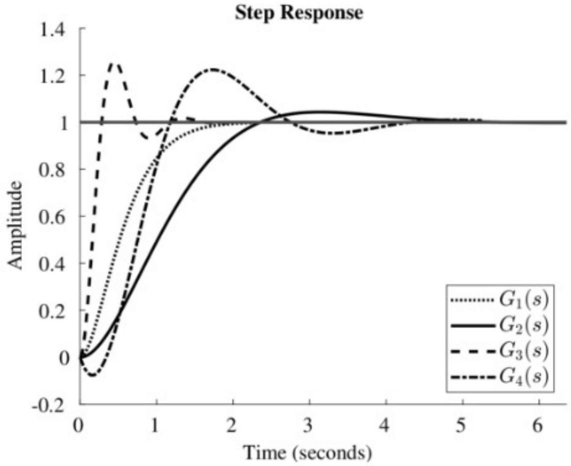


Figure 19: Step reponse plot of four different 2nd order systems. The horizontal line with amplitide equals 1 shows the unit step function.

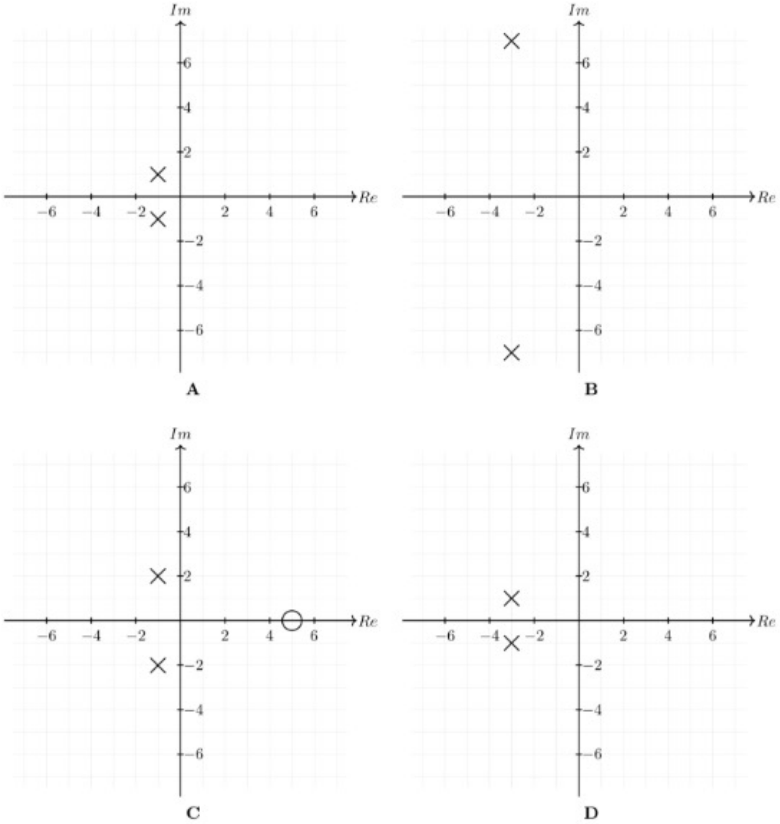


Figure 20: Poles (×) and zeros (○)

Q35 **(1.5 Points)** Mark the correct combinations of poles and zeros plots in Figure 20 with the corresponding 2nd order step response in Figure 19.

Transfer functions	A	B	C	D
$G_1(s)$				x
$G_2(s)$	x			
$G_3(s)$		x		
$G_4(s)$			x	

Summary

Today we learnt:

- Different representation of transfer functions: partial fraction expansion, Root-Locus Form, Bode Form
- How to compute the transfer function graphically
- How poles correspond to exponentials in the time response and determine the system behavior (stability, oscillation, decay rate)
- Zeros act as derivative terms in the system response: they modify the output dynamics without changing system stability
- Min-phase → anticipatory | Non-min-phase → inverse response
- Pole-zero cancellation reduces a pole's effect, but canceling unstable poles makes the system blow up
- Theorems to find a signal's start and steady-state value directly from its Laplace form.

Tips for Problem Set 5

Easy	Medium	Hard
$\mathbf{1, 2, 4}$	$\mathbf{3, 5, 6}$	

- Nr. 1: Use the definitions of poles & zeros, look at the example we did together, look up the cover-up method in the slides
- Nr. 2: Focus on the zeros, look at the problem from the slides
- Nr. 3: think about out pole-zero cancellation in 3b)
- Nr. 4: look at the discriminant i.e. $b^2 - 4ac$ to check if complex or not
- Nr. 5: look at the example we did for the sinusoidal input
- Nr. 6: similar to Nr.5, apply the formula for sinusoidal inputs from the slides

Questions?

Feedback?

Too fast? Too slow? Less theory, more exercises?

I would appreciate your feedback. Please let me know.

https://docs.google.com/forms/d/e/1FAIpQLSdHl0kjWo63aNzDkAV0cnmQadCAj5L0D7v7aSh0BK7BBdEgpA/viewform?usp=header